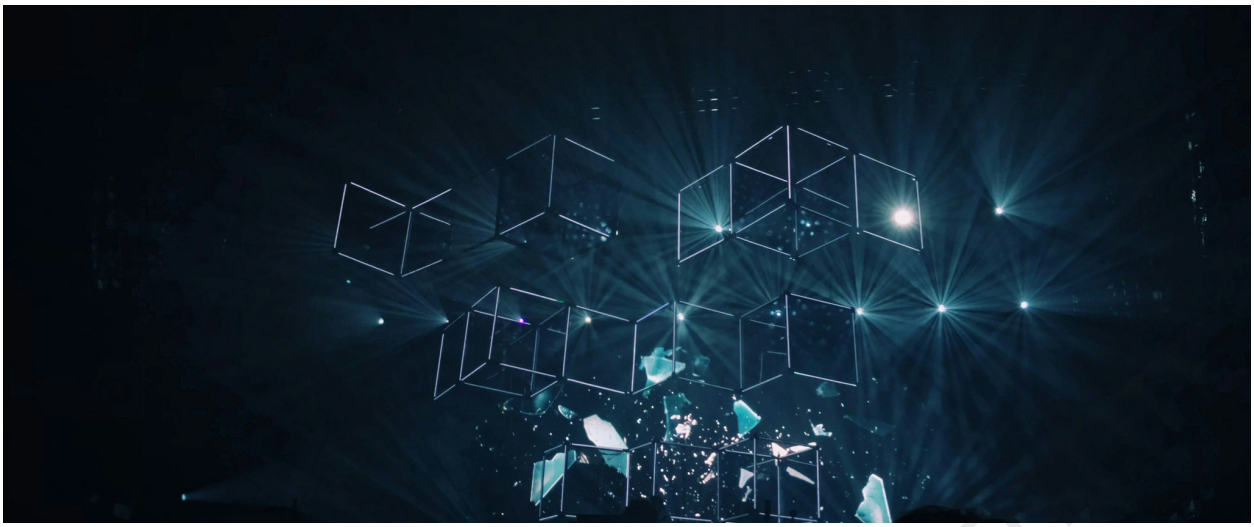

SQL - 1

What is SQL?

SQL (Structured Query Language) is a standard programming language designed for managing and manipulating relational databases. It allows you to store, retrieve, update, and delete data from databases.

What is Data?

Data refers to raw facts, figures, or information that can be processed and stored in a computer system. In databases, data represents real-world entities and their attributes.

Examples of data:

- Names, addresses, phone numbers
- Sales figures, inventory counts
- Student grades, employee records
- Product information, customer details

Types of Data

1. Structured Data

- Organized in a predefined format

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- Stored in tables with rows and columns
- Easy to search and analyze
- Examples: Excel spreadsheets, SQL databases

2. Unstructured Data

- No predefined format or organization
- Cannot be easily stored in traditional databases
- Examples: Text documents, images, videos, social media posts

What is DBMS (Database Management System)?

A DBMS is software that manages databases and provides an interface between users and the database. It handles storage, retrieval, and organization of data.

Key Functions of DBMS:

- Data storage and retrieval
- Data security and access control
- Data backup and recovery
- Concurrent user access management
- Data integrity maintenance

Examples of DBMS:

- File-based systems
- Hierarchical databases
- Network databases
- Object-oriented databases

What is RDBMS (Relational Database Management System)?

RDBMS is a type of DBMS that stores data in tables (relations) with rows and columns. It follows the relational model proposed by E.F. Codd.

Key Features of RDBMS:

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- Data stored in tables with relationships
- Uses SQL for data manipulation
- Supports ACID properties (Atomicity, Consistency, Isolation, Durability)
- Eliminates data redundancy through normalization
- Maintains referential integrity

Popular RDBMS Examples:

- MySQL
- PostgreSQL
- Oracle Database
- Microsoft SQL Server
- SQLite

Key Differences: DBMS vs RDBMS

DBMS	RDBMS
Stores data in files	Stores data in tables
No relationships between data	Supports relationships between tables
Single user access	Multi-user access
Limited security	Enhanced security features
No ACID properties	Supports ACID properties

Basic SQL Components

1. DDL (Data Definition Language)

- CREATE, ALTER, DROP commands
- Define database structure

2. DML (Data Manipulation Language)

- INSERT, UPDATE, DELETE commands
- Manipulate data within tables

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3. DQL (Data Query Language)

- SELECT command
- Retrieve data from the database.

4. DCL (Data Control Language)

- GRANT, REVOKE commands
- Control access permissions

4. TCL (Transaction Control Language)

- COMMIT, ROLLBACK, and SAVEPOINT commands
- Control Transactions

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